Alkane

41. (b) $\underset{\text{Propanoic acid}}{CH_3-CH_2-COOH}+6HI \xrightarrow{\text{Red } P} \underset{\text{Propane}}{CH_3-CH_2-CH_3}+2H_2O+3I_2$

42. (d) $C_2H_5I+2Na+C_3H_7I \xrightarrow[\text{Ether}]{\text{Dry}} \underset{\text{Pentane}}{C_2H_5-C_3H_7}+2NaI$

$C_2H_5I+2Na+C_2H_5I \xrightarrow[\text{Ether}]{\text{Dry}} \underset{\text{Butane}}{C_2H_5-C_2H_5}+2NaI$

$C_3H_7I+2Na+C_3H_7I \xrightarrow[Ether]{Dry} \underset{\text{Hexane}}{C_3H_7-C_3H_7}+2NaI$

43. **(d) Propane** — Propane gives two monobromo isomers (1-bromopropane and 2-bromopropane).

44 **(a) Alkanes** — Kerosene is mainly a mixture of medium-chain alkanes.

45 **(b) Petroleum ether** — On heating petroleum, the lighter fractions (low boiling) vaporize first.

46. **(d) As an antiknock** — Iso-octane increases octane number and reduces knocking.

47. **(c) Petroleum additive** — Tetraethyl lead was used as an antiknock additive in petrol.

48. (d) Cyclohexane, is immiscible and lighter than water. Hence, floats on the surface of water.

49. (a) Methane is the main component of natural gas.

50 **(a) CCl_4 — Carbon tetrachloride is non-inflammable because it contains no hydrogen and does not burn.**

51. **(d) Catalytic hydrogenation — Propene adds H_2 in presence of a catalyst (Ni/Pt) to form propane.**

52. **(a) $(C_2H_5)_4Pb$ — Tetraethyl lead was used as an antiknock agent in petrol.**

53. (a) *Pt./Ni* is used in catalytic reduction of hydrocarbon.

54 **(c) $Br-CH_2-CH_2-Br$ — Ethylene (an alkene) undergoes addition reaction with Br_2 across the double bond to form 1,2-dibromoethane.**

55. (a) Fractional distillation is used because the difference between the boiling point of different component is less.

56. (d) $CH_3 - CH_2 - Cl + \underset{(alc.)}{KOH} \rightarrow \underset{Ethene}{CH_2 = CH_2} + KCl + H_2O$

In presence of alc. *KOH* dehydrohalogenation occur and alkene is formed.

57. (b) Liquefied petroleum gas is a mixture of ethane, propane and butane. The main component is butane.

58. (d) $CH_4 + O_2 \xrightarrow[\text{Limited supply of air}]{\Delta} C + 2H_2O$

It contains 98-99% carbon. It is used in making black ink, paints and shoe polishes.

59. (b) Tetraethyl lead is anti-knocking agent it increases the octane no. of the fuel.

60. (a) *n*-hexane because it is linear therefore strong Vander Waal force.

NEET IIT-JEE BOARD G10 EDUCATIONAL PLATFORM ESTD: 2005